

Appendix C COMPAS

Key references include Angwin et al. (2016); Larson et al. (2016); Dieterich et al. (2016); ProPublica (2016); Equivant (2019); Brennan et al. (2009); Bao et al. (2021); Barenstein (2019).

C.1 Datasheet

C.1.1 Motivation

- **For what purpose was the dataset created?**
This dataset was created for an external audit of racial biases in the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) risk assessment tool developed by Northpointe (now Equivant), which estimates the likelihood of a defendant becoming a recidivist.
- **Who created the dataset and on behalf of which entity?**
The dataset was created by Julia Angwin (senior reporter), Jeff Larson (data editor), Surya Mattu (contributing researcher), Lauren Kirchner (senior reporting fellow). All four contributors were affiliated with ProPublica at the time.
- **Who funded the creation of the dataset?**
The dataset curation work was likely remunerated by ProPublica.

C.1.2 Composition

- **What do the instances that comprise the dataset represent?**
Each instance is a person that was scored for risk of recidivism by the COMPAS system in Broward County, Florida, between 2013–2014. In other words, instances are **defendants**.
- **How many instances are there in total?**
The COMPAS dataset (ProPublica, 2016) consists of **11,757** defendants assessed at the pretrial stage (`compas-scores.csv`). A separate dataset is released for a subset of 7,214 defendants that were observed for two years after screening (`compas-scores-two-years.csv`). Finally a smaller subset of 4,743 defendants focuses on violent recidivism (`compas-scores-two-years-violent.csv`).
- **Does the dataset contain all possible instances or is it a sample of instances from a larger set?**
The dataset represents a **convenience sample** of all individuals that were scored by the COMPAS tool. It concentrates on defendants in Broward County, as it is a large jurisdiction in a state with strong open-records laws (Larson et al., 2016). Moreover, due to Broward County using COMPAS primarily in release/detain decisions prior to a defendant’s trial, scores assessed at parole, probation or other stages were discarded. A notable anomaly in the sample is the low amount of defendants screened between June and July 2013 compared to the remaining time span of the COMPAS dataset (Barenstein, 2019).
- **What data does each instance consist of?**
Instances represent Broward County defendants scored with COMPAS for risk of recidivism. For each defendant the data provided by ProPublica includes tens of variables (~ 50) summarizing their demographics, criminal record, custody and COMPAS scores.
- **Is there a label or target associated with each instance?**
Yes. Instances are associated with two target variables (`is_recid` and `is_violent_recid`), indicating whether defendants were booked in jail with a criminal offense (potentially violent) that took place after their COMPAS screening but within two years. The definition of recidivism and the two-year cutoff were selected by ProPublica staff to align their audit with definitions by Northpointe (Brennan et al., 2009; Angwin et al., 2016).

- **Is any information missing from individual instances?**
Yes. There are several columns where data is missing for one or more instances, including dates when defendants committed the offense (`c_offense_date`) were incarcerated (`c_jail_in`) or released (`c_jail_out`). Missingness in this dataset is not surprising as its curation was a complex endeavour that required cross-referencing information from three separate sources, namely Broward County Sheriff’s Office, Broward County Clerk’s Office and Florida Department of Corrections. Moreover, Northpointe’s response to the ProPublica’s study points out important risk factors considered by the COMPAS algorithm that are not present in the dataset, among which the criminal involvement scale, drug problems sub-scale, age at first adjudication, arrest rate and vocational educational scale (Dieterich et al., 2016). Finally, a clear indication of whether defendants were released or detained pretrial seems to be missing.
- **Are relationships between individual instances made explicit?**
No. While it is plausible for some Broward County defendants to be connected, this information is not available.
- **Are there recommended data splits?**
No.
- **Are there any errors, sources of noise, or redundancies in the dataset?**
Yes. Clerical errors in records caused incorrect matches between individuals’ COMPAS scores and their criminal records, leading to an error rate close to 4% (Larson et al., 2016). Moreover, an important temporal trend was spuriously introduced by ProPublica’s preprocessing in `compas-scores-two-years.csv` and `compas-scores-two-years-violent.csv`, due to which defendants with a screening date after April 2014 are all recidivists (Barenstein, 2019). In terms of redundancies, `compas-scores.csv` contains two identical columns (called `decile_score` and `decile_score.1`).
- **Is the dataset self-contained, or does it link to or otherwise rely on external resources?**
The dataset is **self-contained**.
- **Does the dataset contain data that might be considered confidential?**
No. However it does contain first names and last names of defendants, connecting them to their criminal history.
- **Does the dataset contain data that, if viewed directly, might be offensive, insulting, threatening, or might otherwise cause anxiety?**
Yes. The column `vr_charge_desc` describing violent recidivism charges is one such example.
- **Does the dataset identify any subpopulations (e.g., by age, gender)?**
Yes. The dataset identifies population by age, sex and race. The curators of the COMPAS dataset maintained the race classifications used by the Broward County Sheriff’s Office, identifying individuals as Asian, Black, Hispanic, Native American and White (Larson et al., 2016). Age is reported as an integer, sex as either Male or Female. A distribution along these dimensions is reported in Table 15 which summarizes data in `compas-scores-two-years.csv`. Distributions in remaining files are similar.
- **Is it possible to identify individuals, either directly or indirectly from the dataset?**
Yes. The dataset reports defendants’ first name, last name and date of birth.
- **Does the dataset contain data that might be considered sensitive in any way?**
Yes. The COMPAS dataset reports individuals’ race, criminal history, full name and date of birth.

C.1.3 Collection process

- **How was the data associated with each instance acquired?**
The data was obtained cross-referencing three sources. From the Broward County Sheriff’s Office in Florida, ProPublica obtained COMPAS scores associated with all 18,610 people scored in 2013 and 2014. Defendants’ public criminal records were obtained from the Broward County Clerk’s Office website matching them based on date of birth, first

compas-scores-two-years	
Demographic Characteristic	Values
Percentage of male subjects	80.83%
Percentage of female subjects	19.17%
Percentage of African-American subjects	51.46%
Percentage of Caucasian subjects	33.63%
Percentage of Hispanic subjects	8.67%
Percentage of Asian subjects	0.48%
Percentage of Native American subjects	0.20%
Percentage of people belonging to other races	5.56%
Percentage of people under-19 years old	0.42%
Percentage of people between 20-29 years old	42.41%
Percentage of people between 30-39 years old	28.04%
Percentage of people between 40-49 years old	14.60%
Percentage of people between 50-59 years old	11.00%
Percentage of people between 60-69 years old	3.01%
Percentage of people over-70 years old	0.51%

Table 15: Demographic Characteristics of compas-scores-two-years.

and last names. The dataset was augmented with jail records provided by the Broward County Sheriff’s Office. Finally public incarceration records were downloaded from the Florida Department of Corrections website.

– **What mechanisms or procedures were used to collect the data?**

The original data was plausibly recorded by employees of the Broward County Sheriff’s Office, Broward County Clerk’s Office, and Florida Department of Corrections. The curators of the COMPAS dataset obtained records from the County Sheriff’s Office through a public records request, while data from the County Clerk’s Office and the Florida Department of Correction was downloaded from their official website, matching the methodology of a COMPAS validation study (Larson et al., 2016).

– **If the dataset is a sample from a larger set, what was the sampling strategy?**

In terms of auditing the COMPAS risk assessment tool, this dataset represents a **convenience sample**, focused on a single county and scoring period 2013–2014. Considering a single county in a state with strong open-records laws reduced the data cross-referencing overhead. Concentrating on recent scores predating the study by 2–3 years kept the study timely and permitted a measurement of recidivism aligned with the one by Northpointe. The fact that Northpointe’s response to the ProPublica study only contains minor criticism of the sample (concerning the definition of pretrial defendants (Dieterich et al., 2016)) may be interpreted as testimony to its overall quality. More broadly and beyond the COMPAS audit, arrest data as a proxy for crime brings about specific sampling effects, inevitably mediated by law enforcement practices Xie and Lauritsen (2012); Holmes et al. (2008).

– **Who was involved in the data collection process and how were they compensated?**

The original data was plausibly recorded by Broward County and Florida Department of Corrections employees. On ProPublica’s side, we assume that key curation choices were made and implemented by four employees credited in the article (Angwin et al., 2016) and accompanying technical report (Larson et al., 2016), namely Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner. Given the focus on arrest data, the Broward County law enforcement community is also important in the data sampling process.

- **Over what timeframe was the data collected?**
COMPAS scores are from 2013 and 2014, while jail records cover the period from January 2013 to April 2016. The dataset was first released by ProPublica in May 2016 (ProPublica, 2016).
- **Were any ethical review processes conducted?**
Unknown.
- **Was the data collected from the individuals in question directly, or obtained via third parties or other sources?**
The data was obtained **via third parties**, namely the Broward County Sheriff’s Office in Florida through a public records request, from the Broward County Clerk’s Office through the official website and through the Florida Department of Corrections through the official website. Collection from interested individuals would not have been viable.
- **Were the individuals in question notified about the data collection?**
Likely no. Most of the COMPAS data was publicly available and downloaded from the official websites of Broward County Clerk’s Office and the Florida Department of Corrections.
- **Did the individuals in question consent to the collection and use of their data?**
Likely no. Public availability of arrest/conviction records is associated with collateral consequences that typically damage subjects socially and financially (Pinard, 2010; Angwin et al., 2016).
- **If consent was obtained, were the consenting individuals provided with a mechanism to revoke their consent in the future or for certain uses?**
Likely no.
- **Has an analysis of the potential impact of the dataset and its use on data subjects been conducted?**
Likely no. We are unaware of analyses specifically focused on the COMPAS dataset. More broadly, public availability of criminal records is related to studies on the employability of offenders (Graffam et al., 2008).

C.1.4 Preprocessing/cleaning/labelling

- **Was any preprocessing/cleaning/labeling of the data done?**
Yes. Instances were discarded if assessed with COMPAS at parole, probation or other stages in the criminal justice system. This data is unavailable. Moreover, ProPublica published its datasets with accompanying preprocessing code which has become standard (ProPublica, 2016). The standard preprocessing removes instances for which (1) arrest dates or charge dates are not within 30 days of the COMPAS assessment, (2) true recidivism cannot be decided, (3) charge degree is not defined as misdemeanor or felony, (4) the COMPAS score is not clearly defined. The remaining COMPAS scores were bucketed into low, medium and high risk.
- **Was the “raw” data saved in addition to the preprocessed/cleaned/labeled data?**
Yes. The data is available in the official ProPublica github repository (ProPublica, 2016). This is an intermediate data artifact, already cross-referenced by ProPublica across three separate sources.
- **Is the software used to preprocess/clean/label the instances available?**
Yes. The standard preprocessing software can be found in the official ProPublica github repository (ProPublica, 2016). The software used to cross-reference data from separate sources is not publicly available.

C.1.5 Uses

- **For what tasks has the dataset been used?**
The creators used this dataset to audit the COMPAS tool for racial bias. In the literature it has also been used to evaluate the fairness and accuracy of different algorithms and, more broadly, to study definitions of algorithmic fairness.

- **Is there a repository that links to any or all papers or systems that use the dataset?**
See Appendix A.41 for a (non-exhaustive) list of algorithmic fairness works using this resource.
- **What (other) tasks could the dataset be used for?**
In terms of immediate applications, the dataset could be used to train novel recidivism risk assessment tools. From a methodological perspective, COMPAS may be used in high-stakes domains connected with decision-making about human subjects, including explainable and privacy-preserving ML.
- **Is there anything about the composition of the dataset or the way it was collected and preprocessed/cleaned/labeled that might impact future uses?**
From a very narrow perspective, the fact that all defendants with a screening date after April 2014 are recidivists introduces artificially inflated recidivism base rates (Barenstein, 2019), which would likely be inherited by tools trained on the COMPAS dataset. Moreover, the dataset contains no clear indication concerning pretrial detention or release of defendants. Therefore, researchers must come up with subjective criteria to label individuals as detained or released if they are interested in studying pretrial detention as an intervention deviating from a default course of action (Mishler et al., 2021). From a broader perspective, the data is likely influenced by historical biases in criminal justice, with differential impact on different communities (Holmes et al., 2008; Xie and Lauritsen, 2012; Angwin et al., 2016). Zooming out further, the use of automated risk assessment tools in pretrial decisions is the subject of controversial debate (Barabas et al., 2019) which cannot be overlooked.
- **Are there tasks for which the dataset should not be used?**
Given the above considerations and the narrow geographical scope of the dataset, COMPAS should not be used to train and deploy risk assessment tools for the judicial system. In research settings, users should exercise care in selecting both rows and columns. Bao et al. (2021) suggest avoiding the use of COMPAS to demonstrate novel approaches in algorithmic fairness, as considering data without proper context may bring to misleading conclusions which could misguidedly enter the broader debate on criminal justice.

C.1.6 Distribution

- **Is the dataset distributed to third parties outside of the entity on behalf of which the dataset was created?**
Yes. The COMPAS dataset is publicly available.
- **How is the dataset distributed?**
The dataset is hosted on ProPublica’s official github repository (ProPublica, 2016).
- **When was the dataset distributed?**
Since **May 2016**.
- **Is the dataset distributed under a copyright or other intellectual property (IP) license, and/or under applicable terms of use (ToU)?**
As of June 2021 the COMPAS dataset is freely distributed under ProPublica’s standard ToU (ProPublica, 2021). The dataset cannot be republished in its entirety, it cannot be sold, and can only be used for publication if ProPublica’s work is properly referenced.
- **Have any third parties imposed IP-based or other restrictions on the data associated with the instances?**
Likely no.
- **Do any export controls or other regulatory restrictions apply to the dataset or to individual instances?**
Unknown.

C.1.7 Maintenance

- **Who is supporting/hosting/maintaining the dataset?**
The dataset is currently hosted and maintained by **ProPublica** on github.

-
- **How can the owner/curator/manager of the dataset be contacted?**
The contact for ProPublica’s data store is data.store@propublica.org.
 - **Is there an erratum?**
No. There is no official erratum. An external report highlighting anomalies in the data is available (Barenstein, 2019).
 - **Will the dataset be updated?**
Likely no. In the event of an update, ProPublica’s data store ToU specifies users are solely responsible for checking their sites for updates (ProPublica, 2021)
 - **If the dataset relates to people, are there applicable limits on the retention of the data associated with the instances?**
Unknown.
 - **Will older versions of the dataset continue to be supported/hosted/maintained?**
Unknown.
 - **If others want to extend/augment/build on/contribute to the dataset, is there a mechanism for them to do so?**
Likely no.

C.2 Data Nutrition Label

The following analysis refers to `compas-scores-two-years.csv` after applying the standard COMPAS preprocessing (ProPublica, 2016).

C.2.1 Metadata

METADATA	
Filenames	compas-scores-two-years
Format	csv
Url	https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis
Domain	Law
Keywords	risk assessment, pretrial, recidivism
Type	Tabular
Rows	6,172
Columns	57
% missing cells	5%
Rows with missing cells	100%
License	ProPublica's ToU (ProPublica, 2021)
Released	May 2016
Range	2013-2014 for COMPAS scores, 2013-2016 for arrest and detention history.
Description	Dataset curated by ProPublica to audit COMPAS software for racial biases, focusing on Broward County 2013–2014.

Table 16: Metadata of COMPAS dataset.

C.2.2 Provenance

PROVENANCE	
Source	
Name	Broward County Sheriff's Office
Url	http://www.sheriff.org/
email	//
Name	Broward County Clerk's Office
Url	https://www.browardclerk.org
email	Eclerk@browardclerk.org
Name	Florida Department of Corrections
Url	http://www.dc.state.fl.us/
email	FDCCitizenServices@fdc.myflorida.com
Authors	
Names	Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner
Url	https://www.propublica.org/datastore/dataset/compas-recidivism-risk-score-data-and-analysis
email	data.store@propublica.org

Table 17: Provenance of COMPAS dataset.

C.2.3 Variables

VARIABLES	
id	Unique identifier assigned by the authors
name	Defendant's first and last name
first	Defendant's first name
last	Defendant's last name
compas_screening_date	Day defendant was scored by COMPAS
sex	Defendant's sex
dob	Defendant's date of birth
age	Defendant's age
age_cat	Age quantization: less than 25 25-45 greater than 45
race	Defendant's race: African-American Asian Caucasian Hispanic Native American Other
juv_fel_count	Number of juvenile felonies
decile_score	COMPAS recidivism score (10-point scale)
juv_misd_count	Number of juvenile misdemeanors
juv_other_count	Number of other juvenile convictions (not considering misdemeanor and felonies)
priors_count	Number of prior crimes
days_b_screening_arrest	Days between imprisonment (c_jail_in) and COMPAS screening (compas_screening_date)

Table 18: Variables of COMPAS dataset (1/3).

VARIABLES	
c_jail_in	Date of imprisonment
c_jail_out	Date of release
c_case_number	Alpha-numeric case identifier
c_offense_date	Date on which the offense was committed
c_arrest_date	Date on which defendant was arrested
c_days_from_compas	Days elapsed between offense/arrest and the date of COMPAS screening
c_charge_degree	Degree of charge: F (felony) M (misdemeanor)
c_charge_desc	Textual description of charge
is_recid	Binary indication of recidivism.
r_case_number	Alpha-numeric case identifier for recidivist offense
r_charge_degree	Degree of recidivist charge
r_days_from_arrest	Days elapsed between date of recidivist offense (r_offense_date) and date of recidivist incarceration (r_jail_in)
r_offense_date	Date of recidivist offense
r_charge_desc	Textual description of recidivist charge
r_jail_in	Date of incarceration for recidivist offense
r_jail_out	Date of release for recidivist offense

Table 19: Variables of COMPAS dataset (2/3).

VARIABLES	
violent_recid	Unknown; all nan
is_violent_recid	Binary indication of violent recidivism. If true, then is_recid is true.
vr_case_number	Alpha-numeric case identifier for violent recidivist offense
vr_charge_degree	Degree of violent recidivist offense
vr_offense_date	Date of violent recidivist offense
vr_charge_desc	Textual description of the violent recidivist charge
type_of_assessment	Type of COMPAS assessment - all 'Risk of Recidivism'.
decile_score_1	Identical to decile_score
score_text	Quantization of decile_score: LOW (1-4) MEDIUM (5-7) HIGH (8-10).
screening_date	Identical to compas_screening_date
v_type_of_assessment	Type of COMPAS violent assessment - all 'Risk of Violence'.
v_decile_score	COMPAS violent recidivism score (10-point scale)
v_score_text	Quantization of v_decile_score: LOW (1-4) MEDIUM (5-7) HIGH (8-10).
v_screening_date	Identical to compas_screening_date.
in_custody	Unknown
out_custody	Unknown
priors_count.1	Identical to priors_count.
start	Unknown
end	Unknown
event	Unknown
two_year_recid	Unknown

Table 20: Variables of COMPAS dataset (3/3).

C.2.4 Statistics

STATISTICS						
Ordinal						
name	type	count	uniqueEntries	mostFrequent	leastFrequent	missing
id	int	6,172	6,172	multiple	multiple	0
compas_screening_date	date	6,172	685	2013-04-20	multiple	0
dob	date	6,172	4,830	multiple	multiple	0
age_cat	string	6,172	3	25 - 45	Greater than 45	0
c_jail_in	date	6,172	6,172	multiple	multiple	433
c_jail_out	date	6,172	6,161	2013-09-14 05:58:00	multiple	433
c_offense_date	date	6,172	737	multiple	multiple	1388
c_arrest_date	date	6,172	417	2013-02-06	multiple	8425
r_offense_date	date	6,172	1,041	2014-12-08	multiple	3,182
r_jail_in	date	6,172	928	multiple	multiple	4,175
r_jail_out	date	6,172	893	multiple	multiple	4,175
vr_offense_date	date	6,172	505	2015-08-15	multiple	5,480
v_score_text	string	6,172	3	Low	High	0
v_screening_date	date	6,172	685	2013-04-20	multiple	0
score_text	string	6,172	3	Low	High	0
screening_date	date	6,172	685	2013-04-20	multiple	0
in_custody	date	6,172	1,087	multiple	multiple	0
out_custody	date	6,172	1,097	2020-01-01	multiple	0

Table 21: Ordinal variables statistics of COMPAS dataset

Categorical							
name	type	count	uniqueEntries	mostFrequent	leastFrequent	missing	
name	string	6,172	9,128	mutiple	multiple	0	
first	string	6,172	2,493	michael	multiple	0	
last	string	6,172	3,465	williams	multiple	0	
sex	string	6,172	2	Male	Female	0	
race	string	6,172	6	African-American	Native American	0	
c_case_number	string	6,172	6,172	multiple	multiple	0	
c_charge_desc	string	6,172	390	Battery	multiple	5	
c_charge_degree	string	6,172	2	F	M	0	
r_case_number	string	6,172	2,991	multiple	multiple	3,182	
r_charge_desc	string	6,172	319	Possess Cannabis/ 20 Grams Or Less	multiple	3,228	
r_charge_degree	string	6,172	11	(M1)	(F5)	0	
vr_case_number	string	6,172	693	multiple	multiple	5,480	
vr_charge_desc	string	6,172	82	Battery	multiple	5,480	
vr_charge_degree	string	6,172	10	(M1)	(F5)	5,480	
type_of_assessment	string	6,172	1	Risk of Recidivism	Risk of Recidivism	0	
v_type_of_assessment	string	6,172	1	Risk of Violence	Risk of Violence	0	
is_recid	binary	6,172	2	0	1	0	
is_violent_recid	binary	6,172	2	0	1	0	
event	binary	6,172	2	0	1	0	
two_year_recid	binary	6,172	2	0	1	0	

Table 22: Categorical variables statistics of COMPAS dataset

Quantitative									
name	type	count	min	median	max	mean	stdDev	miss	zeros
age	int	6,172	18	31	96	34.53	11.73	0	0
juv_fel_count	int	6,172	0	0	20	0.06	0.46	0	5,964
juv_misd_count	int	6,172	0	0	13	0.09	0.50	0	5,820
juv_other_count	int	6,172	0	0	9	0.11	0.47	0	5,711
priors_count	int	6,172	0	1	38	3.25	4.74	0	2,085
days_b_screening_arrest	int	6,172	-30.0	-1	30.0	-1.74	5.08	0	1,379
c_days_from_compas	int	6,172	0	1	9,485	24.90	276.81	0	869
r_days_from_arrest	int	6,172	-1	0	993	20.10	76.54	4,175	1,452
decile_score	int	6,172	1	4	10	4.42	2.84	0	0
v_decile_score	int	6,172	1	3	10	3.64	2.49	0	0
start	int	6,172	0	0	937	13.32	50.14	0	3,485
end	int	6,172	0	539	1,186	555.05	400.26	0	1

Table 23: Quantitative variables statistics of COMPAS dataset.